

Biochemistry of dietary, functional and structural Lipids



BY

Dr. Sally Mohammed El-Hefnawy

Assistant professor of Medical Biochemistry & Molecular Biology

By the end of this lecture, the student will be able to:

a1. Enumerate importance of lipids.

a2 . List classification of lipids.

a3. Define essential fatty acids and List their classification.

a4. Define prostaglandins and their role.

a5. Define rancidity, its types and causes.

a6. Describe compound lipids and its importance.

a7. Define steroids and their types.



- **Triacylglycerols** are quantitatively the most important class of dietary fat.
- Consumption of saturated fats is positively associated with high levels of total plasma cholesterol and an increased risk of coronary heart diseases.
- Unsaturated fatty acids are generally derived from vegetables and fish.

- Monounsaturated fats lower total plasma cholesterol.
- Diets rich in olive oil (high in monounsaturated oleic acid), show a low incidence of coronary heart diseases.
- Dietary ω -3 polyunsaturated fats (fish oil) as linolenic acid suppress cardiac arrhythmias, reduce serum triacylglycerols, decrease the tendency for thrombosis, lower blood pressure, and substantially reduce risk of cardiovascular mortality.

Importance of lipids

- (1) An efficient source of energy (1 gm of fat $\longrightarrow$ about 9 cal).
- (2) It serves as thermal insulators in the subcutaneous tissues and around certain organs e.g kidney
- (3) Lipids act as electrical insulators, allowing rapid propagation of the impulses along the myelinated nerves.
- (4) Source for Fat-soluble vitamins and essential fatty acids.
- (6) Cholesterol and its important derivatives as bile acids and steroid hormones.

Ester bond - COO -

Fatty acid + Alcohol $\longrightarrow$ ester + H₂O

R-COOH + R-OH $\longrightarrow$ R-COO-R + H₂O

Classification of lipids

(A) Simple lipids: These are esters of fatty acids with alcohol.

According to the alcohol, they are sub-classified into:

(1) Neutral fats (TAG): These are esters of fatty acids with glycerol.

(2) Waxes: These are esters of fatty acids with higher molecular weight alcohols.

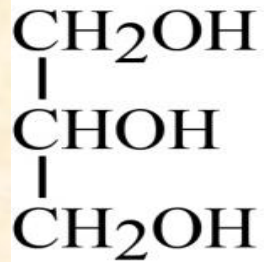
(B) Complex or compound lipids: These are esters of fatty acids with alcohol containing in addition other groups. According to these groups, they are sub-classified into: Phospholipids Glycolipids -Proteolipids -sulfolipids

(C) Derives lipids: as steroids.

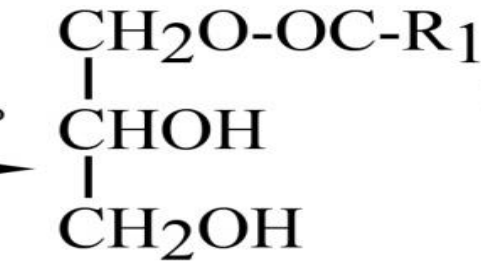
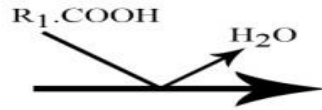
Glycerol

Glycerol is a trihydric alcohol which can be esterified with one two or three fatty acids giving mono-, di-, or triglycerides respectively.

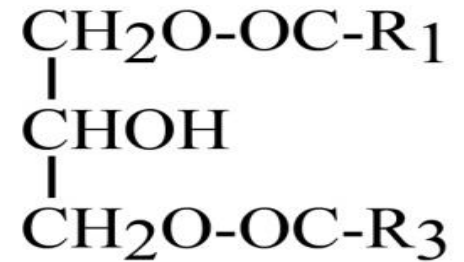
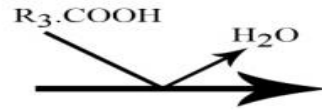
Glycerol is always phosphorylated at C-3 by the enzyme glycerokinase giving **glycerol- 3 phosphate**.



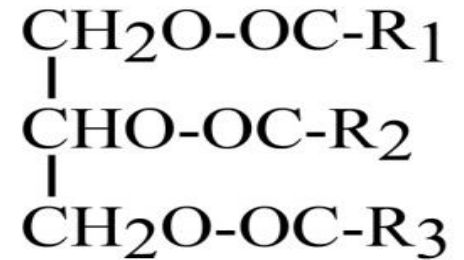
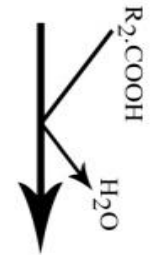
Glycerol



Monoglyceride



1,3-diglyceride



Triglyceride

Fatty acids

Fatty acids (FA) are monocarboxylic acids (COOH), which occur mainly as esters in natural fats and oils. Fatty acids may be saturated or unsaturated

(A) Saturated fatty acids (containing no double bonds)

- (1) Short chain FA (2-10 C)
- (2) Long chain FA (more than 10 C)

(B) Unsaturated fatty acids: (containing one or more double bonds).

- (1) Monounsaturated fatty acids include: palmitoleic (C16) and oleic acid (C18).
- (2) Polyunsaturated fatty acids include: Linoleic acid (C18)& Linolenic acid (C18) and arachidonic acid (C20).

Table 6.1 Saturated fatty acids commonly found in natural fats

Acid	Formula	Carbon atoms
Acetic	CH_3COOH	2
Propionic	$\text{C}_2\text{H}_5\text{COOH}$	3
Butyric	$\text{C}_3\text{H}_7\text{COOH}$	4
Caproic	$\text{C}_5\text{H}_{11}\text{COOH}$	6
Caprylic	$\text{C}_7\text{H}_{15}\text{COOH}$	8
Decanoic	$\text{C}_9\text{H}_{19}\text{COOH}$	10
Lauroic	$\text{C}_{11}\text{H}_{23}\text{COOH}$	12
Myristic	$\text{C}_{13}\text{H}_{27}\text{COOH}$	14
Palmitic	$\text{C}_{15}\text{H}_{31}\text{COOH}$	16
Stearic	$\text{C}_{17}\text{H}_{35}\text{COOH}$	18
Arachidic	$\text{C}_{19}\text{H}_{39}\text{COOH}$	20
Behenic	$\text{C}_{21}\text{H}_{43}\text{COOH}$	22
Lignoceric	$\text{C}_{23}\text{H}_{47}\text{COOH}$	24

Linoleic acid & Linolenic and arachidonic acid cannot be synthesized in the body at a sufficient rate and must be taken in diet. They are called essential fatty acids.

Sources of essential fatty acids

- Plant oils e.g. corn oil, sunflower oil and linseed oil.



- Fish oils e.g., cod liver oil, which contain ω -3 polyunsaturated fatty acids





Linoleic acid (C_{18:2})



α -Linolenic acid (C_{18:3})



Arachidonic acid (C_{20:4})

Importance of essential fatty acids:

1- Formation of phospholipids and cholesterol ester.

2- Enter in structure of cell membrane and are required for the fluidity of cell membrane.

3-They protect against atherosclerosis and coronary heart disease.

4-Arachidonic acid is a precursor of eicosanoids (e.g. prostaglandins).

Prostaglandins (PGs)

- PGs (one of eicosanoids) as they are derived from arachidonic acid containing 20 carbon atoms (eicosa) by the action of Cyclooxygenase enzyme (COX).
- They act as local hormones on muscles, blood vessels and adipose tissues causing contraction of smooth muscles and control of blood pressure.

Rancidity

Definition: Rancidity is the development of bad odor and taste in the fat. It is caused by its exposure to moisture, oxygen, warmth, light and certain metals as copper and lead.

Types of rancidity:

There are three types:

(a) Hydrolytic rancidity:

It is caused by bacterial enzymes which cause hydrolysis of fat liberating free fatty acids. It is caused by increased water content of fats as in butter.

(b) Ketonic rancidity:

This type of rancidity is caused by the action of certain fungi, producing ketones of bad odor and taste.

(c) Oxidative rancidity:

- Exposure of lipids to oxygen (autooxidation) forms peroxides which break into short chain alcohols and acids having bad odor.
- Peroxidation of lipids is responsible for the deterioration of food and also for damage to the tissues in vivo causing cancer, inflammatory disease, atherosclerosis and aging.
- Oxidative rancidity can be prevented by anti-oxidants.

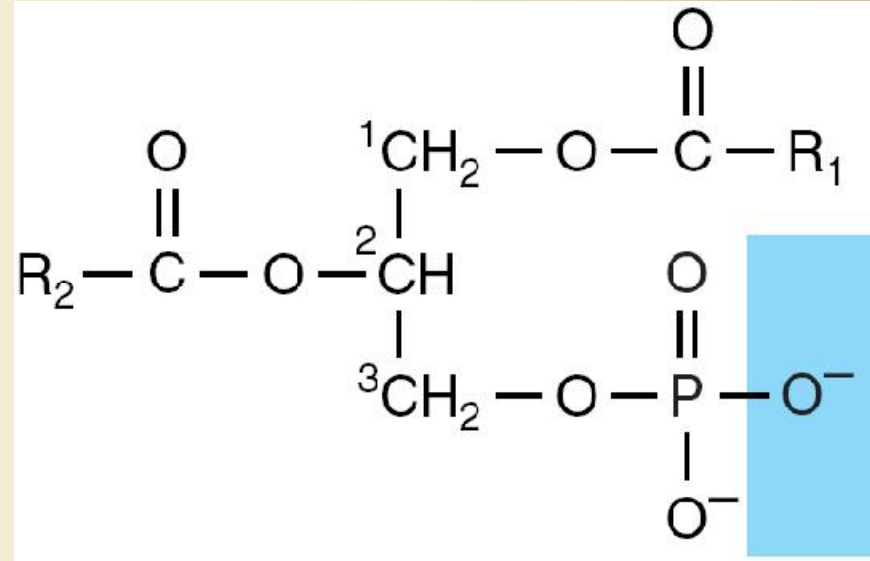
Compound Lipids

e.g. Phospholipids

(1) Phosphatidic acid:

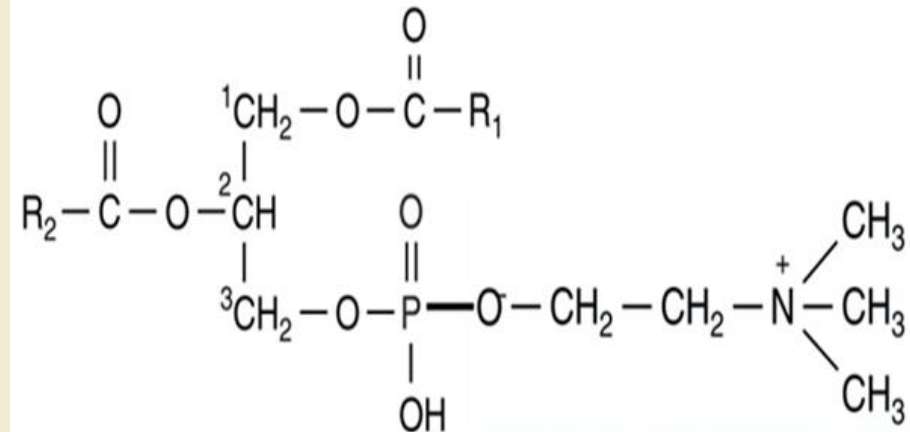
It consists of glycerol esterified with 2 FA, one of them is usually saturated at the 1 position and the other FA is usually unsaturated at the 2 position. It contains phosphoric acid at the third position.

Phosphatidic acid is important as an intermediate in the synthesis of triacylglycerol.



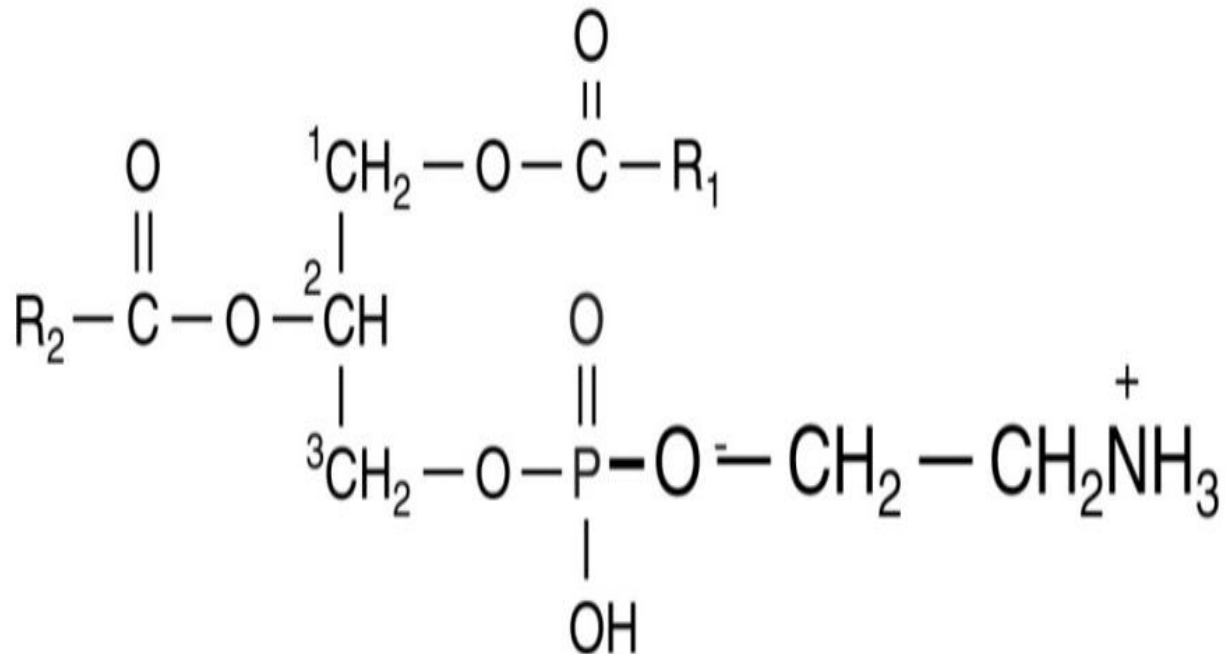
(2) Phosphatidyl cholines (Lecithins):

- ✓ Lecithins consist of phosphatidic acid esterified with choline
- ✓ They are the most important phospholipids of the cell membrane
- ✓ Dipalmitoyl-lecithin (lung surfactant) is a very effective surface active agent, preventing adherence of the inner surfaces of the lungs. It is continuously secreted by alveolus forming a monolayer over the watery surface of the alveolus causing decrease of surface tension during expiration and inspiration. It is absent from the lungs of premature infants causing **respiratory distress syndrome**.



(3) Phosphatidyl ethanolamine (Cephalins):

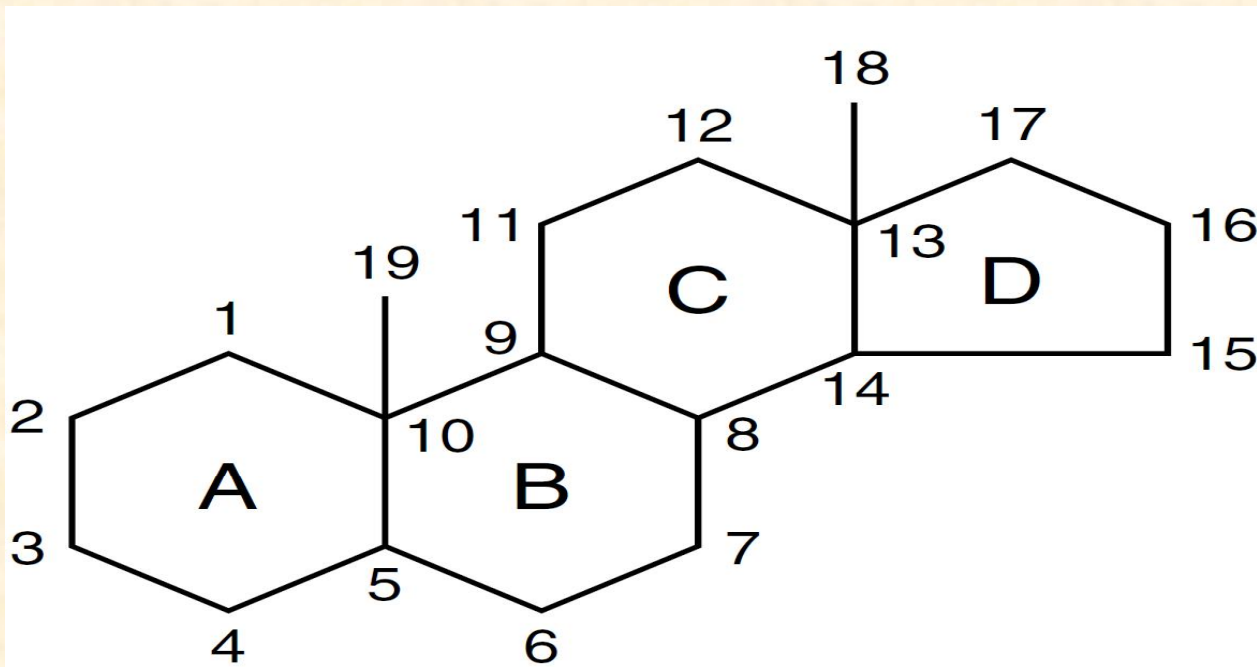
Cephalins consist of phosphatidic acid esterified with ethanolamine. It is present mainly in myelin sheath of nerves and liver cell membrane.



Derived Lipids

STEROIDS

-Cyclopentanoperhydrophenanthrene nucleus is composed of 4 rings: A, B, C and D.



• The steroid nucleus.

Steroids include 4 groups :

1- Sterols : cholesterol which is major constituent of plasma membrane. Cholesterol in blood and tissues is found in combination with fatty acids as cholesteryl esters.

2- Bile acids: There are 2 types: cholic acid and chenodeoxycholic acid. These are synthesized in the liver from cholesterol.

3- Steroid hormones:

(A) Sex hormones: estrogens, androgens and progesterone.

(B) Glucocorticoids (cortisone) and mineralocorticoids (aldosterone).

3- Vitamin D : synthesized in the skin from cholesterol and activated in liver and kidney.

1- The number of double bonds in Arachidonic acid

- a) 1
- b) 2
- c) 3
- d) 4

2- The precursor for vitamin D is:

- (a) Cholesterol
- (b) Arachidonic acid
- (c) Triglycerol
- (d) Phospholipid

3- ----- is an essential fatty acid:

- (a) Stearic acid
- (b) Palmitoleic acid
- (c) Linoleic acid
- (d) Oleic acid

4- The development of bad odor and taste in fat describes the processes of:

a- hydrogenation of fats

b- Saponification

c- rancidity

d- denaturation

5- Prostaglandins are synthesized in the body from:

- (a) Cholesterol
- (b) Stearic acid
- (c) Arachidonic acid
- (d) Sphingosine

Reference

1- Harpers illustrated Biochemistry

2- Lippincott illustrated reviews integrated systems

https://alryadamy.sharepoint.com/:v:/g/personal/sally_alhefnawy_rst_edu_eg/EVg6ZXj8x9dFmUHpKxsnHfgBdvMRzbG_F1UrGraKx5IsLQ?e=bePIt1&nav=eyJyZWZlcnJhbEluZm8iOmsicmVmZXJyYWxBcHAiOiJTdHJlYW1XZWJBcHAiLCJyZWZlcnJhbFZpZXciOiJTdGFyZURpYWxvZyIsInJlZmVycmFsQXBwUGxhdGZvcn0iOiJXZW5iLCJyZWZlcnJhbE1vZGUiOiJ2aWV3In19



القلب

Thank you

Best Wishes